

Part 1: Equations of Motion - Simulation Model

Problem Description

A point mass m is connected to a massless rod of length ℓ . The rod is connected to a wall with a nonlinear spring and a damper. The system is subject to gravity g .

Physical Parameters

- Gravity: $g = 9.8 \text{ m/s}^2$
- Rod length: $\ell = 0.25 \text{ m}$
- Mass: $m = 0.1 \text{ kg}$
- Nonlinear spring constants: $k_1 = 0.02, k_2 = 0.01$
- Viscous friction (damping): $b = 0.1$
- Input torque: τ (limited to $\pm 3.0 \text{ N-m}$)

Configuration Variable

- $q = \theta$ (angle of the rod from the horizontal, as shown in the diagram)

Tasks

1. Find the kinetic energy K .
2. Find the potential energy P .
3. Find the Lagrangian $L = K - P$.
4. Find the generalized forces.
5. Derive the equations of motion using Euler-Lagrange.